

Heat and Thermodynamics

Topic of heat and thermodynamics

1. Thermodynamic system (Definition)

2. 1st Law of Thermodynamics

3. Application of 1st Law

$$C_p - C_v = R$$

$$PV^\gamma = \text{constant}$$

4. 2nd Law of Thermodynamics (Definition)

5. Heat Engine and Efficiency.

6. Carnot Heat Engine

$$\rightarrow \eta = 1 - \frac{Q_2}{Q_1} ; \eta = 1 - \frac{T_2}{T_1}$$

7. Entropy

Change in Reversible process

Change in irreversible process

8. 3rd Law of Thermodynamics.

Thermodynamic System:

Systems are classified as follows:

Isolated System:

If there is no interchange of energy or matter between the system and surroundings, then the system is called isolated system.

Closed system:

If no energy or matter crosses the boundary, then the system is called closed system.

Open system:

If there is an interchange of energy or matter between the system and surroundings, then the system is called open system.

Thermodynamic process:

A thermodynamic process is the energetic development of a thermodynamic system proceeding from an initial state to a final state.

Different thermodynamic processes are:

Isochoric process:

If during a process, the volume is constant then the process is called isochoric process.

Isobaric process:

If during a process, the pressure is constant then the process is called isobaric process.

Isothermal Process:

If during a process, the temperature is constant then the process is called isothermal process.

Adiabatic process:

A process is called adiabatic process if it is carried by a system enclosed by an adiabatic boundary; such that no flow of heat is possible across the boundary.

Reversible process:

A reversible process can be defined as one whose direction can be reversed by an infinitesimal change of in some property of the system.

Irreversible process: If there is a finite temperature difference between system and surroundings the direction of the heat flow can't be reversed by an infinitesimal change in temperature of the system and the process is irreversible.

First law of Thermodynamics:

"When heat is supplied to any system then a part of that heat helps to increase the internal energy of the system and the remaining part of the energy is used by the system to do external work on the environment."

Explanation:

Let us consider,

U_1 be the internal energy of a system. If Q amount of heat is added to the system then the internal energy of the system becomes U_2 .

The change in internal energy can then be denoted as,

$$\Delta U = U_2 - U_1 \text{ ————— ①}$$

Thus the first law of the thermodynamics gives,

$$Q = \Delta U + W \text{ ————— ②}$$

Equation ② is the mathematical form of the first law of thermodynamics.

For an infinitesimal reversible process, the first law takes the form,

$$\boxed{dQ = dU + dW.}$$

Application of First law of Thermodynamics:

Molar Specific Heat:

The amount of heat needed to increase the temperature one Kelvin of one mole gas is called molar specific heat.

$$C = \frac{\Delta Q}{m\Delta T} \text{ or, } C = \frac{dQ}{mdT}$$

Molar Specific Heat at Constant Pressure:

At constant pressure, the amount of heat needed to increase the temperature one Kelvin of one mole gas is called molar specific heat at constant pressure.

$$C_p = \frac{dQ}{mdT}$$

Molar Specific Heat at Constant Volume:

At constant volume, the amount of heat needed to increase the temperature one Kelvin of one mole gas is called molar specific heat at constant volume.

$$C_v = \frac{dQ}{mdT}$$

Relation between C_p and C_v

Let us consider a cylinder containing one mole gas. A frictionless piston is attached to the cylinder. Let p be the pressure, V be the volume, T be the temperature and U be the internal energy of the gas.

If dQ amount of heat is given to the cylinder then the increase in internal energy, volume and temperature and dU , dV and dT respectively and dW is the external work then from the first law of thermodynamics, we get,

$$dQ = dU + dW$$

$$\text{or, } dQ = dU + PdV \quad \text{--- (1)}$$

But increase in internal energy dU per mole gas is equal to the product of molecular specific heat at constant volume, C_v and increase in temperature dT .

$$dU = C_v dT \quad \text{--- (2)}$$

We know that, the amount of heat needed to increase the temperature one kelvin of one mole gas is called molar specific heat at constant pressure.

$$C_p = \frac{dQ}{dT}$$

$$\Rightarrow dQ = C_p dT \text{ --- (3)}$$

If R is the molar gas constant, then for one mole gas we get,

$$PV = RT$$

Differentiating with respect to T ,

$$\frac{d}{dT}(PV) = \frac{d}{dT}(RT)$$

$$\Rightarrow P \frac{dV}{dT} = R$$

$$\Rightarrow PdV = RdT \text{ --- (4)}$$

Putting the values of dU , dQ , and PdV from equation (2), (3) and (4) into equation (1). We get,

$$C_p dT = C_v dT + RdT$$

$$\Rightarrow C_p = C_v + R$$

$$\Rightarrow C_p - C_v = R$$

Here molar gas constant R is universal constant and is a positive quantity. So C_p is always greater than C_v .

In case of Adiabatic change for an Ideal Gas $PV^\gamma = \text{constant}$

Let us consider a cylinder containing one mole gas. A frictionless piston is attached to the cylinder. Let P be the pressure, V be the volume, T be the temperature and U be the internal energy of the gas.

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$$dQ = dU + dW$$

$$\Rightarrow dQ = dU + PdV \quad \text{--- (1)}$$

But increase in internal energy dU per mole gas is equal to the product of molecular specific heat at constant volume, C_v and increase in temperature dT .

$$dU = C_v dT$$

Now from equation (1)

$$dQ = C_v dT + PdV$$

At Adiabatic process, $dQ = 0$

$$C_v dT + PdV = 0 \quad \text{--- (2)}$$

Again we know for one mole gas,

$$PV = RT$$

Differentiating with respect to T ,

$$\frac{d}{dT}(P) \frac{d}{dT}(PV) = \frac{d}{dT}(RT)$$

$$\text{or, } P \frac{dV}{dT} + V \frac{dP}{dT} = R$$

$$\text{or, } PdV + VdP = R dT$$

$$\Rightarrow dT = \frac{PdV + VdP}{R} \quad \text{--- (3)}$$

Substituting equation (3) in equation (2), we get,

$$C_v \left(\frac{PdV + VdP}{R} \right) + PdV = 0$$

$$\Rightarrow C_v PdV + C_v VdP + R PdV = 0$$

$$\Rightarrow C_v PdV + C_v VdP + (C_p - C_v) PdV = 0$$

$$\Rightarrow C_v VdP + C_p PdV = 0$$

$$\Rightarrow VdP + \left(\frac{C_p}{C_v} \right) PdV = 0$$

$$\Rightarrow VdP + \gamma PdV = 0 \quad \left[\text{where } \gamma = \frac{C_p}{C_v} \right]$$

$$\therefore \frac{dP}{P} + \gamma \frac{dV}{V} = 0$$

Integrating above equation, we get,

$$\int \frac{dp}{p} + \gamma \int \frac{dv}{v} = \text{constant}$$

$$\Rightarrow \ln p + \gamma \ln v = \text{constant}$$

$$\Rightarrow \ln p + \ln v^{\gamma} = \text{constant}$$

$$\Rightarrow \ln(pv^{\gamma}) = \text{constant}$$

$$\Rightarrow pv^{\gamma} = \text{constant}$$

$$\therefore pv^{\gamma} = \text{constant}$$

Second Law of Thermodynamics:

Different scientist have stated the law in different forms:

Carnot's statement:

No engine can be built which can extract a fixed amount of heat and will convert totally into work.

Clausius's statement:

It is impossible for a self acting machine, unaided by external agency, convey heat from one body at a lower temperature to another body at a higher temperature.

Planck's statement:

It is impossible to construct an engine which can extract heat continuously from a source of heat and completely transforms into work.

Kelvin's statement:

Continuous flow of energy can't be obtained from an object cooling it than the coolest part of its surroundings.

Heat Engine and Efficiency of Heat Engine:

Heat Engine:

Any mechanism for the conversion of heat into mechanical power is called a heat engine or any suitable device which can convert heat into mechanical work is called heat engine.

Efficiency of Heat Engine:

The efficiency of heat engine is defined as the ratio of work done during a cycle to the heat absorbed during the cycle.

Thus if ω be the amount of work obtained from heat engine in one cycle at the expense of amount of heat, then its efficiency η is defined as,

$$\eta = \frac{\omega}{Q}$$

$$\Rightarrow (Q_1 - Q_2) / Q_1$$

$$= 1 - \frac{Q_2}{Q_1}$$

Carnot's Heat Engine and It's Efficiency:

Carnot's Theorem:

(a) No engine can be more efficient than a perfectly reversible engine working between the same two temperatures.

(b) The efficiency of all reversible engines, working between the same two temperatures is the same, whatever be the working substance.

A Carnot engine is a hypothetical engine that operates on a Carnot cycle. The basic model for this engine was given by Sadi Carnot in 1824.

The cycle carried through a Carnot cycle is the prototype for all heat engines. The feature that is common to all heat engines is that, they receive an input of heat at one temperature, do mechanical work on their surroundings, and reject heat at some lower temperature.

When any working substance is carried through a cyclic process, there is no change in its internal energy in any

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Carnot's Heat Engine and It's Efficiency:

A Carnot heat engine is a hypothetical engine that operates on the reversible Carnot cycle. The basic model for this engine was developed by Sadi Carnot in 1824.

A system carried through a Carnot cycle is the prototype of all cyclic heat engines. The feature that is common to all such devices is, that, they receive an input of heat at one or more higher temperature, do mechanical work on their surroundings and reject at some lower temperature.

When any working substance is carried through a cyclic process, there is no change in its internal energy in any

complete cycle and from the first law the net flow of heat Q into the substance in any complete cycle is equal to the workdone w by the engine per cycle.

$$w = Q_1 - Q_2 = Q$$

where, Q_1 is ~~that~~ the heat flowing in the system and Q_2 is the heat flowing out of the system per cycle.

The thermal efficiency η of heat engine is defined as

$$\eta = \frac{\text{work output}}{\text{heat input}}$$

$$= \frac{\text{heat converted into work per cycle}}{\text{heat drawn from the heat source into the system per cycle.}}$$

$$= \frac{w}{Q_1}$$

$$= \frac{Q_1 - Q_2}{Q_1}$$

$$= 1 - \frac{Q_2}{Q_1} \quad \text{--- (1)}$$

For Carnot's engine the heat flowing in or out of the system is proportional to the temperature of the heat reservoir.

$$Q \propto T \\ \Rightarrow \frac{Q}{T} = \text{constant}$$

If T_1 and T_2 are the temperature of the source and sink respectively, then

$$\frac{Q_1}{T_1} = \frac{Q_2}{T_2} \\ \Rightarrow \frac{Q_2}{Q_1} = \frac{T_2}{T_1}$$

~~If T_1 and T_2 are the temperature of the source~~
Then from equation (1), we get.

$$\eta = 1 - \frac{Q_2}{Q_1} \\ = 1 - \frac{T_2}{T_1} \\ = (T_1 - T_2) / T_1$$

But efficiency is expressed in term of percentage.
So, $\eta = \{(T_1 - T_2) / T_1\} \times 100\%$.

Entropy:

The disability of converting energy of a system or unavailability of energy to convert into work is entropy.

Explanation:

It has not yet been possible to measure the absolute magnitude of entropy. If any body absorbs or rejects heat then its entropy is changed.

The change of entropy is measured by the rate of absorption or rejection of heat by the system with respect to temperature.

If any a system absorbs or rejects dQ amount of heat at temperature T , then the change of entropy

$$dS = \frac{dQ}{T}$$

Change of Entropy:

- Change of Entropy in Reversible Adiabatic process:

By definition, the heat absorbed in a reversible adiabatic process is zero,

$$dQ_r = 0$$

Hence for such a process entropy change is given by,

$$dS = \frac{dQ_r}{T} = 0$$

$$\therefore S = \text{constant}$$

Thus the entropy of a reversible adiabatic process is constant and such process is called isentropic.

- Change of Entropy in Reversible Isothermal process:

In a reversible isothermal process the temperature remains constant and for such a process,

$$S = \int \frac{dQ}{T}$$

$$= \frac{1}{T} \int dQ$$

$$= \frac{Q_r}{T}$$

That is, the change of entropy in a reversible isothermal process is equal to the heat absorbed or given out by the system dividing by the thermodynamic temperature of the system

If the heat flows into the system Q is positive and entropy increases. If the heat flows out of the system Q is negative and entropy decreases.

- Change of Entropy in irreversible process:

Let two objects completely isolated from the environment are in contact with each other. The temperatures of two objects are respectively T_1 and T_2 . If $T_1 > T_2$ then heat will flow from the warm object to the cold object. Let within a very short period of time dQ amount of heat is transferred from the warm one to the cold one. That is the warm object losses dQ amount of heat and the cold object gains dQ amount of heat.

Therefore,

$$\frac{-dQ}{T_1} = \text{Decrease of entropy of the warm object}$$

And, $\frac{dQ}{T_2} = \text{Increase of entropy of the cold object.}$

Therefore, the change of entropy of the system,

$$dS = -\frac{dQ}{T_1} + \frac{dQ}{T_2}$$

It is observed from the above equation that,

$$\text{Since, } T_1 > T_2$$

$$\therefore dS > 0$$

That is the change of entropy is always positive.

The Third Law of Thermodynamics:

"The entropy of a system approaches a constant value as its temperature approaches absolute zero."

Example-4.1

A motor car tyre has a pressure of 2 atmospheres at the room temperature of 27°C . If the tyre suddenly bursts, find the resulting temperature.

Solution:

Here,

$$P_1 = 2 \text{ atmospheres}$$

$$T_1 = 273 + 27 = 300\text{K}$$

$$P_2 = 1 \text{ atmospheres}$$

$$T_2 = ?$$

$$\gamma = 1.4$$

We know that,

$$\frac{P_1^{\gamma-1}}{T_1^{\gamma}} = \frac{P_2^{\gamma-1}}{T_2^{\gamma}}$$

$$\Rightarrow \left(\frac{P_2}{P_1}\right)^{\gamma-1} = \left(\frac{T_2}{T_1}\right)^{\gamma}$$

$$\Rightarrow \left(\frac{1}{2}\right)^{0.4} = \left(\frac{T_2}{300}\right)^{1.4}$$

$$\Rightarrow 0.4 \log(0.5) = 1.4 [\log T_2 - \log 300]$$

$$\Rightarrow -0.1204 = 1.4 \log T_2 - 3.4680$$

$$\Rightarrow 1.4 \log T_2 = 3.4680 - 0.1204$$

$$\Rightarrow 1.4 \log T_2 = 3.3476$$

$$\Rightarrow \log T_2 = \frac{3.3476}{1.4} = 2.3911$$

$$\Rightarrow T_2 = 10^{2.3911} / \text{antilog}(2.3911)$$

$$\Rightarrow T_2 = 246.1 \text{ K}$$

$$\therefore T_2 = -26.9^\circ \text{C}$$

Example-4.2

A quantity of air at 27°C and atmospheric pressure is suddenly compressed to half its original volume. Find the final (i) pressure and (ii) temperature. (Given $\gamma = 1.4$, $2^{1.4} = 2.64$)

Solution:

$$(i) P_1 = 1 \text{ atmosphere}$$

$$P_2 = ?$$

$$\gamma = 1.4$$

$$V_1 = V, V_2 = \frac{V}{2}$$

During sudden compression, the process is adiabatic.

$$P_1 V_1^\gamma = P_2 V_2^\gamma$$

$$\begin{aligned}
 \Rightarrow P_2 &= P_1 \left[\frac{V_1}{V_2} \right]^\gamma \\
 &= 1 [2]^{1.4} \\
 &= \text{Antilog}[1.4 \log 2] \\
 &= \text{Antilog}[1.4 \times 0.30] \\
 &= 2.636 \text{ atmospheres.}
 \end{aligned}$$

$$\textcircled{\text{ii}} V_1 = V \quad \gamma = 1.4$$

$$V_2 = \frac{V}{2}$$

$$T_1 = 300 \text{ K}$$

$$T_2 = ?$$

$$\begin{aligned}
 T_1 (V_1)^{\gamma-1} &= T_2 (V_2)^{\gamma-1} \\
 &= T_2 [2]^{1.4-1} \\
 &= 300 \times [2]^{0.4} \\
 &= 395.9 \text{ K}
 \end{aligned}$$

$$\therefore T_2 = 122.85^\circ \text{C}$$

Example-4.3

Air at N.T.P is compressed adiabatically to half of its volume. Calculate the change in its temperature.

Solution:

Let the initial Temperature be T_1 K and the final temperature be T_2 K.

Initial Volume = V_1

Final Volume = $V_2 = \frac{V_1}{2}$

During an adiabatic process,

$$T_1 V_1^{\delta-1} = T_2 V_2^{\delta-1}$$

$$\Rightarrow T_2 = T_1 \left[\frac{V_1}{V_2} \right]^{\delta-1}$$

$$\Rightarrow T_2 = T_1 [2]^{\delta-1}$$

But δ for air = 1.4

$$\Rightarrow T_2 = T_1 [2]^{1.4-1}$$

$$\Rightarrow T_2 = 1.319 T_1$$

$\therefore$ Change in temperature = $T_2 - T_1$

$$= 1.319 T_1 - T_1$$

$$= 0.319 T_1 \text{ K}$$

Example-4.4

1 g of a molecule of a monoatomic ($\gamma = 5/3$) perfect gas at 27°C is adiabatically compressed in a reversible process from an initial pressure of 1 atmosphere to a final pressure of 50 atmospheres. Calculate the resulting difference in temperature.

Solution:

In a reversible adiabatic process,

$$\frac{P_1^{\gamma-1}}{T_1^{\gamma}} = \frac{P_2^{\gamma-1}}{T_2^{\gamma}}$$

$$\Rightarrow \left(\frac{P_2}{P_1}\right)^{\gamma-1} = \left(\frac{T_2}{T_1}\right)^{\gamma}$$

Hence, $P_2 = 50$

$$P_1 = 1$$

$$T_1 = 273 + 27 = 300\text{K}$$

$$T_2 = ?$$

$$\gamma = \frac{5}{3}$$

$$\therefore \left(\frac{50}{1}\right)^{2/3} = \left(\frac{T_2}{300}\right)^{5/3}$$

$$\Rightarrow \frac{2}{3} \log(50) = \frac{5}{3} [\log T_2 - \log 300]$$

$$\Rightarrow T_2 = 1434 \text{ K}$$

$$= 1161^\circ\text{C}$$

Ans:

Example-4.5

A quantity of dry air at 27°C is compressed (i) slowly and (ii) suddenly to $\frac{1}{3}$ of its volume. Find the change in temperature in each case, assuming γ to be 1.4 for dry air.

Solution:

1. When the process is slow, the temperature of the system remains constant. Therefore, there is no change in temperature.
2. When the compression is sudden, the process is adiabatic.

Here, $V_1 = V$, $V_2 = \frac{V}{3}$, $T_1 = 300 \text{ K}$, $T_2 = ?$, $\gamma = 1.4$

$$T_2(V_2)^{\gamma-1} = T_1(V_1)^{\gamma-1}$$

$$\Rightarrow T_2 = T_1 \left(\frac{V_1}{V_2} \right)^{\gamma-1}$$

$$= 300 \left[\frac{3V}{V} \right]^{\gamma-1}$$

$$= 300 [3]^{1.4-1}$$

$$= 465.5 \text{ K}$$

$$= 192.5^\circ\text{C}$$

The temperature of air increases by,

$$= 192.5 - 27$$

$$= 165.5^{\circ}\text{C}$$

$$= 165.5\text{K}$$

Ans:

Example-4.6

Find the efficiency of the Carnot's engine working between the steam point and the ice point.

Solution:

$$\text{Here, } T_1 = 273 + 100 = 373\text{K}$$

$$T_2 = 273 + 0 = 273\text{K}$$

$$\eta = 1 - \frac{T_2}{T_1}$$

$$= 1 - \frac{273}{373}$$

$$= 0.268$$

$$\therefore \text{Efficiency} = 0.268 \times 100$$

$$= 26.8\%$$

Ans:

Example-4.7

Find the efficiency of a Carnot's engine working between 127°C and 27°C . It absorbs 80 cal of heat. How much heat is rejected.

Solution:

$$T_1 = 273 + 127 = 400\text{K}$$

$$T_2 = 273 + 27 = 300\text{K}$$

$$\eta = 1 - \frac{T_2}{T_1}$$
$$= 1 - \frac{300}{400}$$

$$= 0.25$$

$$= 0.25 \times 100$$

$$= 25\%$$

$$W = \eta Q_1 = 0.25 \times 80 = 20 \text{ cal}$$

$$\therefore \text{Heat rejected } Q_2 = Q_1 - W$$

$$= 80 - 20$$

$$= 60 \text{ cal}$$

Ans:

Example-4.8

A Carnot's engine whose temperature of the source is 400 K takes 200 calories of heat at this temperature and rejects 150 calories of heat to the sink. What is the temperature of the sink? Also calculate the efficiency of the engine.

Solution:

$$Q_1 = 200 \text{ cal} ; Q_2 = 150 \text{ cal}$$

$$T_1 = 400 \text{ K} ; T_2 = ?$$

$$\frac{Q_1}{T_1} = \frac{Q_2}{T_2}$$

$$\Rightarrow T_2 = \frac{Q_2}{Q_1} \times T_1$$

$$= \frac{150}{200} \times 400$$

$$= 300 \text{ K}$$

$$\text{Efficiency, } \eta = 1 - \frac{T_2}{T_1}$$

$$= 1 - \frac{300}{400}$$

$$= 0.25$$

$$\therefore \text{Efficiency} = 0.25 \times 100$$
$$= 25\%$$

Ans:

Example-4.9

A Carnot's engine working as a refrigerator between 260K and 300K receives 500 calories of heat from the reservoir at the lower temperature. Calculate the amount of heat rejected to the reservoir at the higher temperature. Calculate also the amount of work done in each cycle to operate the refrigerator. (1 calorie = 4.2 joules).

Solution:

$$Q_1 = ?$$

$$Q_2 = 500 \text{ cal}$$

$$T_1 = 300 \text{ K}$$

$$T_2 = 260 \text{ K}$$

$$\frac{Q_1}{Q_2} = \frac{T_1}{T_2}$$

$$\Rightarrow Q_1 = \frac{T_1}{T_2} \times Q_2$$

$$= \frac{300}{260} \times 500$$

$$= 576.92 \text{ cal}$$

$$\text{Work done, } W = Q_1 - Q_2$$

$$= 76.92 \text{ cal}$$

$$= 76.92 \times 4.2 = 323.08 \text{ Joules}$$

Example-4.10

A Carnot's engine whose lower temperature heat-sink is at 27°C has its efficiency 40%. What is the temperature of the heat sources. By how much should the temperature of the source be raised, if the efficiency is to be raised to 70%.

Solution:

$$T_2 = 273 + 27 = 300\text{K}$$

$$\eta_1 = \frac{40}{100} = 0.4$$

$$\eta_2 = \frac{70}{100} = 0.7$$

① Temperature of the source:

$$\eta_1 = 1 - \frac{T_2}{T_1}$$

$$\Rightarrow \frac{T_2}{T_1} = 1 - \eta_1$$

$$\Rightarrow T_1 = \frac{T_2}{1 - \eta_1} = \frac{300}{1 - 0.4} = \frac{300}{0.6} = 500\text{K}$$

$$= 227^{\circ}\text{C}$$

② Required rise in source Temperature:

$$T_1 = \frac{T_2}{1 - \eta_2}$$
$$= \frac{300}{1 - 0.7} = 1000\text{K} = 727^\circ\text{C}$$

∴ Required rise in the source temperature,

$$= T_2' - T_1$$
$$= 1000 - 500$$
$$= 500\text{K} = 500^\circ\text{C}$$

Example-4.12

A Carnot engine whose low temperature reservoir is at 7°C has an efficiency of 50%. It is desired to increase the efficiency at 70%. By how many degrees should the temperature of the high temperature reservoir be increased?

Solution:

In the first case,

$$\eta = 50\% = 0.5$$

$$T_2 = 273 + 7 = 280\text{K}$$

$$T_1 = ?$$

$$\eta = 1 - \frac{T_2}{T_1}$$

$$\Rightarrow 0.5 = 1 - \frac{280}{T_1}$$

$$\Rightarrow T_1 = 560 \text{ K}$$

In the second case,

$$\eta' = 70\% = 0.7$$

$$T_2 = 280 \text{ K}$$

$$T_1' = ?$$

$$\eta' = 1 - \frac{T_2}{T_1'}$$

$$\Rightarrow 0.7 = 1 - \frac{280}{T_1'} \Rightarrow T_1' = 933.33 \text{ K}$$

$$\Rightarrow T_1' = 840 \text{ K}$$

$$\therefore \text{Increase in Temperature} = 933.33 - 560 = 373.33 \text{ K}$$

Example - 4.21

Ans!

An inventor claims to have developed an engine working between 600 K and 300 K capable of having an efficiency of 52%. Comment on his claim.

Solution: $T_1 = 600 \text{ K}, T_2 = 300 \text{ K}$

$$\eta = 1 - \frac{T_2}{T_1}$$

$$= 1 - \frac{300}{600} = \frac{1}{2} = 50\%$$

The efficiency claimed = 52%.

It means that the efficiency of the engine is more than the efficiency of a Carnot's engine working between the same two temperature limits. But no engine can have an efficiency more than a Carnot's engine, so his claim is invalid.

Example-4.22

A Carnot engine absorbs 10^4 calories of heat from a reservoir at 627°C and rejects heat to a sink at 27°C . What is its efficiency? How much work does it perform?

[J = 4.2 J/calories]

Solution :

$$T_1 = 627 + 273 = 900\text{K}$$

$$T_2 = 27 + 273 = 300\text{K}$$

$$\eta = 1 - \frac{T_2}{T_1}$$

$$= 1 - \frac{300}{900}$$

$$= \frac{2}{3}$$

$$= 66.67\%$$

$$\text{Also, } \frac{Q_1}{Q_2} = \frac{T_1}{T_2}$$

$$\Rightarrow Q_1 = 10^4 \text{ calories}$$

$$\therefore Q_2 = \frac{Q_1 \times T_2}{T_1}$$

$$= \frac{10^4 \times 300}{900}$$

$$= \frac{10^4}{3} \text{ calories}$$

$$\text{Work done, } W = Q_1 - Q_2$$

$$\Rightarrow W = 10^4 - \left(\frac{10^4}{3}\right)$$

$$= \frac{2 \times 10^4}{3} \text{ calories}$$

$$= \frac{2 \times 10^4}{3} \times 4.2$$

$$= 2.8 \times 10^4 \text{ J}$$

Ans:

Example - 4.25

A Carnot engine has its source at 100°C and its sink is maintained at a constant temperature by means of ice at 0°C . If it is working at the rate of 100 watts, how much ice will melt in one minute?

Solution:

$$P = 100 \text{ watts} = 100 \text{ J/s}$$

$$\text{Work done in one minute} = 100 \times 60 = 6000 \text{ J}$$

$$T_1 = 100^{\circ}\text{C} = 100 + 273 = 373 \text{ K}$$

$$T_2 = 0^{\circ}\text{C} = 0 + 273 = 273 \text{ K}$$

$$\frac{Q_1 - Q_2}{Q_1} = \frac{T_1 - T_2}{T_1}$$

$$\frac{\omega}{Q_1} = \frac{T_1 - T_2}{T_1} \quad \text{--- (1)}$$

$$\frac{Q_1}{Q_2} = \frac{T_1}{T_2} \quad \text{--- (2)}$$

From (1) and (2)

$$\frac{\omega}{Q_1} = \frac{T_1 - T_2}{T_1}$$

$$\Rightarrow Q_2 = \left[\frac{\omega T_2}{T_1 - T_2} \right]$$

$$= \frac{6000 \times 272}{100} = 16320 \text{ J}$$

Suppose, m kg of ice melts

$$Q_2 = mL$$

$$\Rightarrow m = \frac{Q_2}{L} = \frac{16320}{80 \times 10^3 \times 4.2} = 0.0485 \text{ kg}$$

Ans:

Example-4.27

Calculate the rise in temperature of a gas initially at 27°C , if its pressure is suddenly doubled, $\gamma = 1.4$

Solution:

$$\frac{P_1^{\gamma-1}}{T_1^{\gamma}} = \frac{P_2^{\gamma-1}}{T_2^{\gamma}}$$

Hence, $\left(\frac{P_2}{P_1}\right)^{\gamma-1} = \left(\frac{T_2}{T_1}\right)^{\gamma}$

Hence,

$$P_2 = 2P_1$$

$$T_1 = 27^\circ\text{C} = 27 + 273 \\ = 300 \text{ K}$$

$$T_2 = ?$$

$$\gamma = 1.4$$

$$\therefore (2)^{1.4-1} = \left(\frac{T_2}{300}\right)^{1.4}$$

$$\Rightarrow 0.4 \log 2 = 1.4 [\log T_2 - \log 300]$$

$$\Rightarrow 0.1204 = 1.4 \log T_2 - 3.4679$$

$$\Rightarrow 1.4 \log T_2 = 3.4679 - 0.1204$$

$$\Rightarrow \log T_2 = 2.39112$$

$$\Rightarrow T_2 = \text{Antilog}(2.39112)$$

$$\Rightarrow T_2 = 201.89 \text{ K}$$

$$\text{Rise in Temperature} = (201.89 -$$